Cyber Security in Agriculture: Protecting Smart Farms from Digital Attacks

A Research Perspective with a Story — **Presented by Mohan A V | ID No: AMB2139**

> "Imagine a farm where tractors drive themselves, sensors talk to the cloud, and irrigation listens to weather apps… but one day, a hacker decides to 'water' the crops 24/7. Welcome to the world of smart farming—and its hidden risks!"

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Problem Statement

Modern agriculture relies on IoT, AI, and cloud systems — but **security is often ignored**. Cyber threats now target the very systems that feed the world.

Hacked Irrigation

Smart irrigation systems can be remotely exploited to waste or withhold water.

Stolen Crop Data

Sensitive yield and soil data can be stolen or manipulated by bad actors.

Hijacked Tractors

Automated tractors can be remotely controlled, causing physical damage.

Lack of Awareness

Farmers largely lack cybersecurity knowledge to defend their systems.

> 😂 "Traditional pests eat crops… Cyber pests eat your **data, money, and control systems!**"

Research Question: How can we secure smart agricultural systems from cyber threats while maintaining efficiency?

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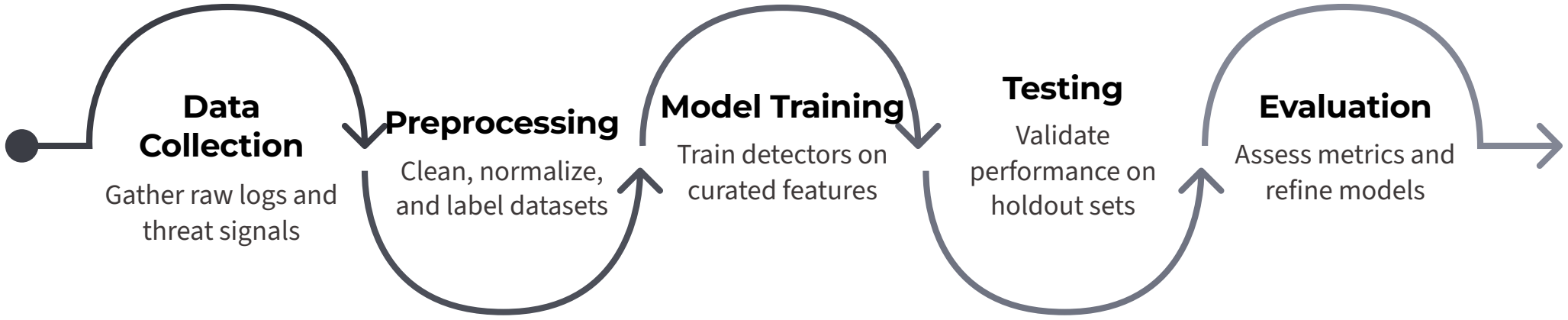
Methodology

Our research acts like a **"digital scarecrow"** — not to scare birds, but to detect hackers before they damage the farm. The approach follows a structured pipeline from vulnerability discovery to model evaluation.



The methodology identifies vulnerabilities in smart farming systems, simulates cyber-attacks such as data tampering and unauthorised access, applies machine learning models for threat detection, and evaluates performance based on accuracy and detection rate.

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Data Description

Data Sources

- IoT sensor data (soil moisture, temperature, humidity)
- Network traffic logs from smart devices
- User access logs

Key Features

- Sensor readings & Device IDs
- Login attempts
- Data transmission patterns

Sample Data Snapshot

Soil Moisture	Temperature	Login Attempts	Status
45%	28°C	2	Normal
90%	30°C	15	Suspicious

🌧️ "Too much water is bad for crops… Too many login attempts are bad for systems!"

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Model & Approach

Like a farmer recognising weeds among crops, ML models detect **unusual patterns — "cyber weeds"** — hidden within data streams.

ML Techniques

Random Forest

Detects anomalies in sensor and access data.

SVM

Classifies normal behaviour vs. active attacks.

Neural Networks

Advanced detection of complex threat patterns.

Security Methods

Encryption

Protects data in transit and at rest.

Authentication

Verifies user and device identities.

IDS

Intrusion Detection Systems monitor for breaches.

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Results

Our "digital scarecrow" successfully caught the hacker before the crops were flooded! 😲 Key findings demonstrate that machine learning delivers robust, real-world protection for smart farms.

85–95%

Detection Accuracy

ML models detect cyber-attacks with high accuracy across test scenarios.

↓ Damage

Early Detection

Timely alerts significantly reduce the impact of successful intrusions.

High

IoT Vulnerability

IoT systems are highly exposed to attack without active protection layers.

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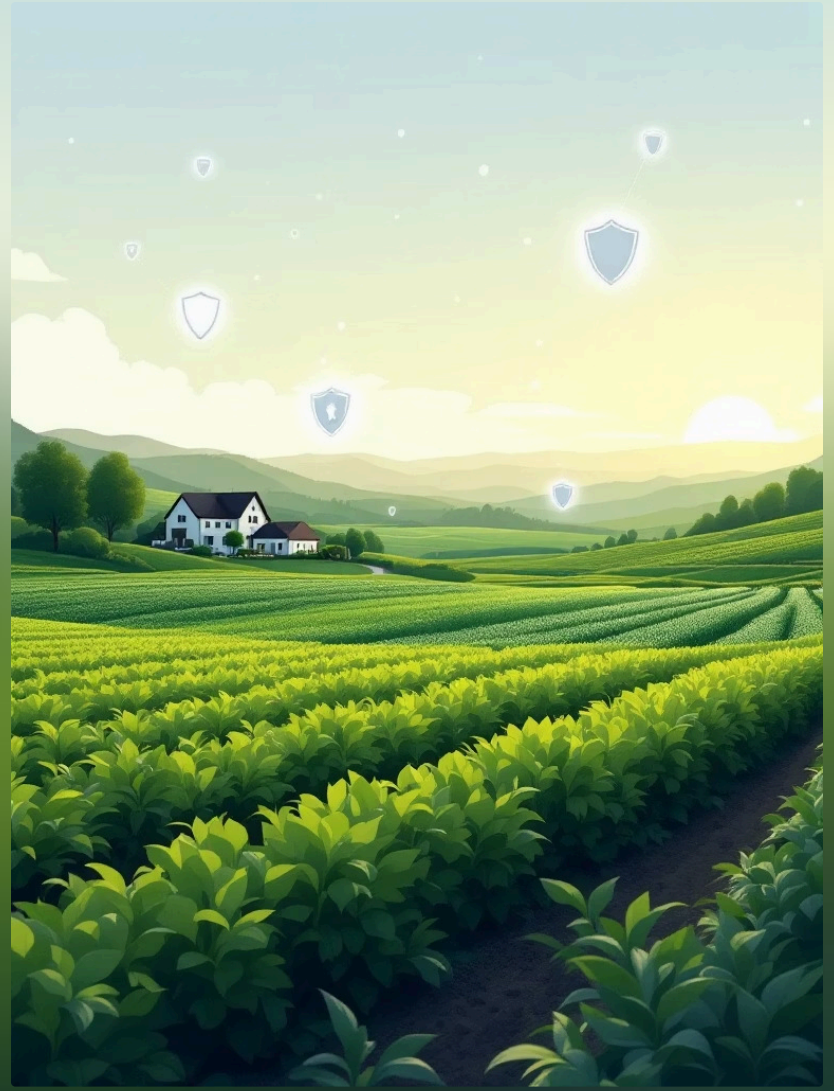

Conclusion & Future Scope

Key Takeaways

- Smart farming increases efficiency but introduces serious cyber risks
- Machine learning effectively detects and prevents attacks
- Cybersecurity must be integrated into agriculture systems from the ground up

Future Directions

- AI-based real-time monitoring
- Blockchain for secure data sharing
- Farmer awareness and training programmes

🌾 "Just like crops need pesticides, smart farms need **cyber protection tools** to survive."

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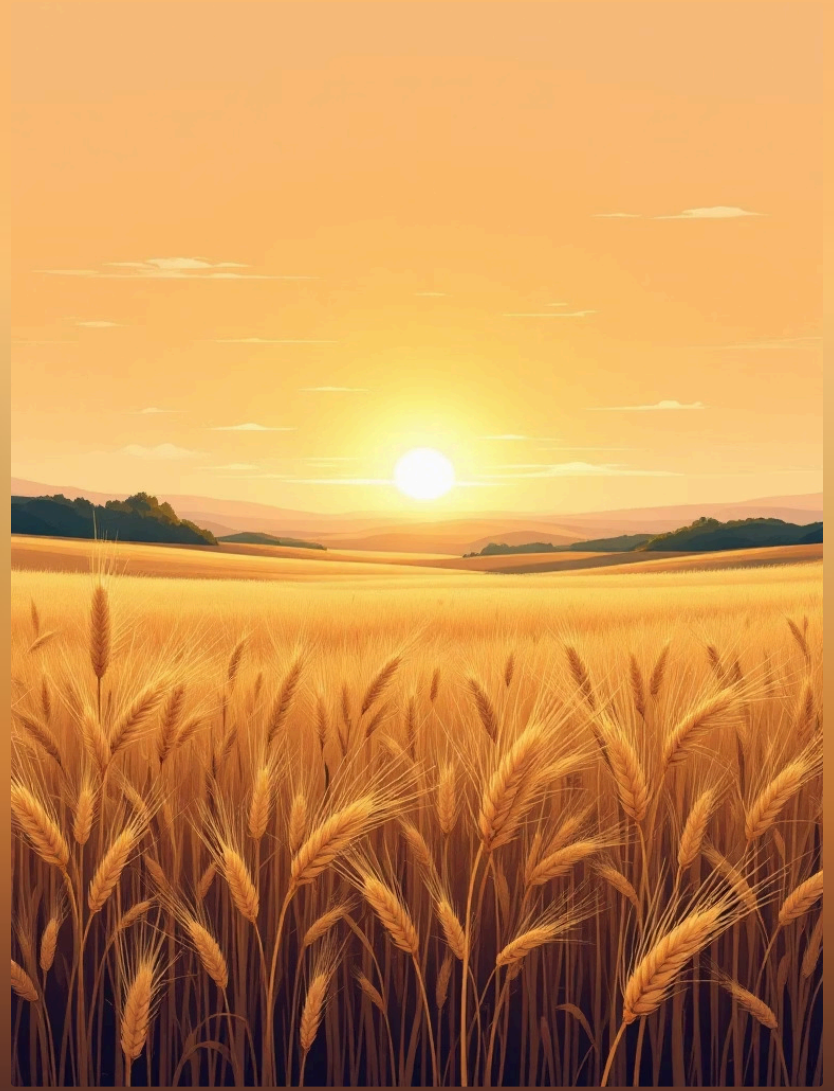

Thank You

Questions & Discussion

We welcome your questions, feedback, and ideas. Let's work together to build a safer, smarter future for agriculture.

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CYBER SECURITY IN AGRICULTURE

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